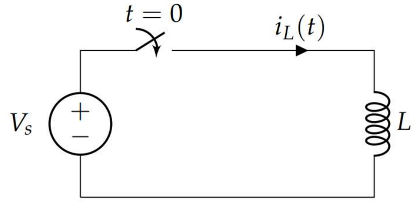
Assignment 1

torsdag 16. januar 2025 14:28

Problem 1 (Inductors and capacitors)

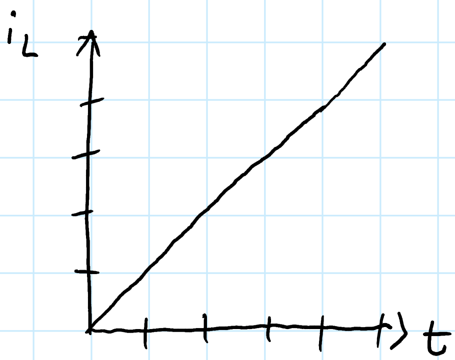
(a) In the circuit on the right, $V_s = 1\,\mathrm{V}$ and $L = 1\,\mathrm{H}$. Before $t = 0$ the current through the inductor is equal to zero. At $t = 0$ the circuit is closed. Set up the differential equation for the circuit when $t \geq 0$, then calculate the current $i_L(t)$. Sketch its graph for $0 < t < 5\,\mathrm{s}$.



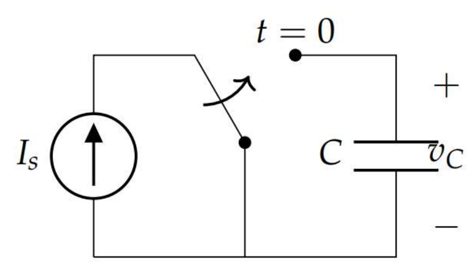
$$V_L = L\frac{di}{dt} \Rightarrow \frac{V_L}{L} = \frac{di}{dt}, \quad V_L = V_S = 1\,V$$

$$\int \frac{di}{dt}\,dt = \int \frac{V_L}{L}\,dt$$

$$i_L(t) = \frac{V_L}{L}t + i(0) \overset{i_0=0}{=} t\ A$$



(b) The circuit on the right has a constant current source I_s. Before $t = 0$ the capacitor is energy-less, i.e., $v_C(t < 0) = 0$. At $t = 0$ the switch is flipped, so that the current is forced through the capacitor C. Find the voltage $v_C(t)$ for $I_s = 1\,\mathrm{A}$ and $C = 1\,\mathrm{F}$. Sketch its graph for $0 < t < 5\,\mathrm{s}$.

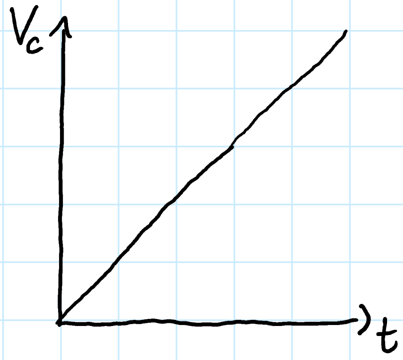


$$i_c = C\frac{dV_c}{dt} \Rightarrow \frac{i_s}{C}dt = dV_c$$

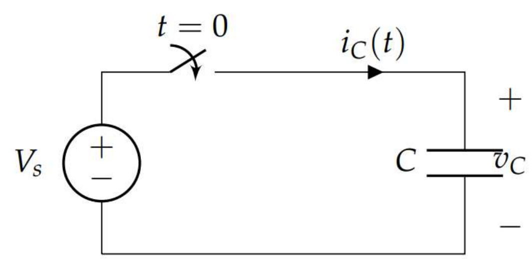
$$\int 1\,dV_c = \int \frac{i_s}{C}\,dt$$

$$V_c(t) = \frac{i_s}{C}t + V_c(0) \overset{V_0=0}{=} t\ V$$

$$V_c(t) = \frac{i_s}{C} t + V_c(0) = t \ \text{V}$$



(c) Now, we will study a somewhat unrealistic circuit in order to illustrate how much a capacitor resists the change of voltage over it. The circuit on the right is closed at $t = 0$, and $V_s = 1\,\text{V}$ and $C = 1\,\text{F}$. Moreover, $v_C(t < 0) = 0$. Find the current $i_C(t)$. This is a mystical calculation. Discuss what would happen in practice.



$$i_c = C \frac{dV_c}{dt}$$

$$V_s = V_c = 1\text{V} \quad \text{for } t \geq 0$$

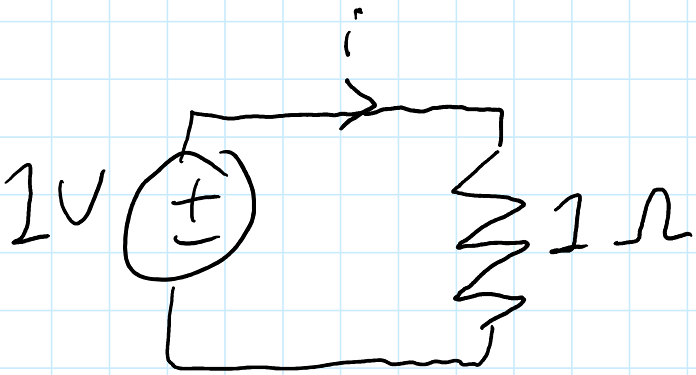
$$\frac{dV_c}{dt} = 0 \quad \text{because } V_c \text{ is const.}$$

$$\Rightarrow i_c = 0 \quad \text{for } t > 0$$

$$\text{At } t=0 \quad i_c = \frac{dV_c}{dt} = \infty$$

In practice we should see a a big (but n infinite) current spike before it goes to 0.

d) Stationy equivalent:

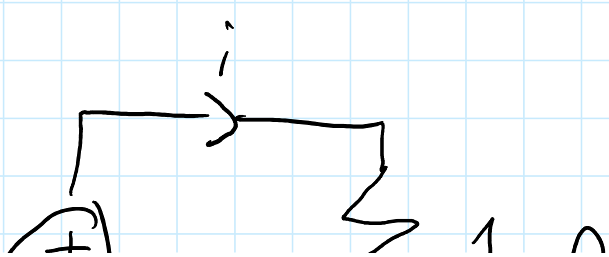


$$V = RI \Rightarrow I = \frac{V}{R} = 1A$$

e) Equivalent to open circuit:

$i = 0\,A$

d)



$i = 1A$

Problem 2

a) $i_L(t) = \frac{V_L}{L} t + i(0) = \frac{16V}{100mH} \cdot t = 160 \cdot t \ A$

for $0 \leq t \leq 10ms$

b) $i_L(\frac{1}{100}) = 160 \cdot \frac{1}{100} A = 1{,}6 \ A$

$10ms \leq t \leq T$:

$i_L(\frac{1}{100}) = 1{,}6A = \frac{-5V}{100mH} \cdot \frac{1}{100} + C \ A$

$C = 2{,}1$

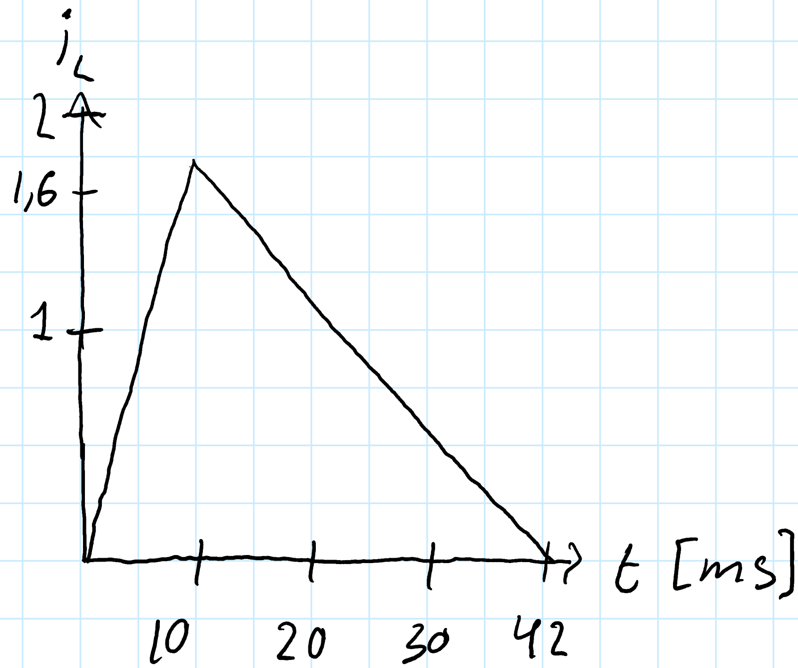
$\Rightarrow i_L(t) = -50 \frac{V}{H} \cdot t + 2{,}1 A$

$i_L(T) = -50 \frac{V}{H} \cdot t + 2{,}1 A = 0$

2,1

$$T = \frac{2,1}{50} s = 0,042 s = 42 ms$$

c)



Problem 3

a/b
$$V_s = V_R + V_L$$

$$V_s = R i_L + L \frac{di_L}{dt}$$

$$i_L(t) = C e^{-\frac{R}{L}t} + \frac{V_s}{R}$$

$$i_L(0) = C + \frac{V_s}{R} = 0 \Rightarrow C = -\frac{V_s}{R}$$

$$i_L(t) = \frac{V_s}{R}\left[1 - e^{-\frac{R}{L}t}\right]$$

$$i_L(\infty) = \frac{V_s}{R}$$

c) $i_L(0^-) = 0$ because it's an open circuit

$i_L(0^+) = 0$ because current lags

voltage / inductors resist change in current

d) $i_L(t) = \frac{V_S}{R}\left[1 - e^{-\frac{R}{L}t}\right]$

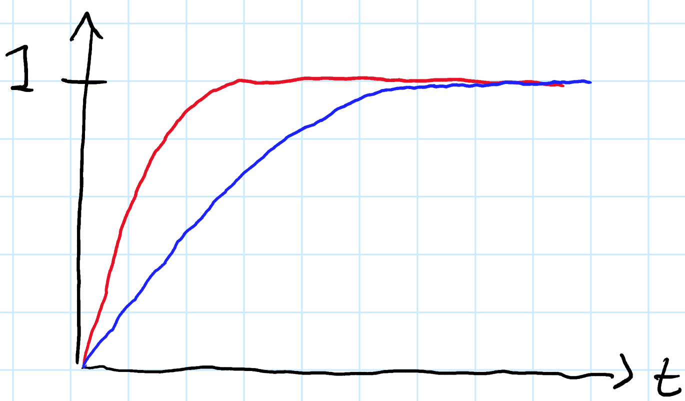
$V_S = 1V \quad R = 1\Omega \quad L = 0{,}02H$ or $0{,}2H$

$L = 0{,}02H$:

$i_L(t) = 1 - e^{-50t}$

$L = 0{,}2H$

$i_L(t) = 1 - e^{-5t}$



The current changes more slowly with more Henry. Both expressions converge to $\frac{V_S}{R}$.

Problem 4

a) $V_L = L\frac{di_L}{dt}$

Using $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$:

$$\mathcal{L}\{V_L\} = \mathcal{L}\left\{L\frac{di_L}{dt}\right\}$$

$$V_L(s) = sL\,I_L(s) - L\,i_L(0)$$

b) $V_s = R\cdot i_L + L\frac{di_L}{dt}$

$$\frac{V_s}{s} = R\,I_L(s) + sL\,I_L(s) \underbrace{- L\,i_L(0)}_{0}$$

$$I_L(s) = \frac{\frac{V_s}{s}}{R + sL}$$

$$i_L(t) = \frac{V_s}{L}\cdot\mathcal{L}^{-1}\left\{\frac{1}{s(s+\frac{R}{L})}\right\}$$

Partial fractions:

$$\frac{1}{s(s+\frac{R}{L})} = \frac{\frac{L}{R}}{s} - \frac{\frac{L}{R}}{s+\frac{R}{L}}$$

$$\mathcal{L}\left\{\frac{\frac{L}{R}}{s}\right\} = \frac{L}{R} \quad \text{and} \quad \mathcal{L}\left\{-\frac{\frac{L}{R}}{s+\frac{R}{L}}\right\} = -\frac{L}{R}e^{-\frac{R}{L}t}$$

$$i_L(t) = \frac{V_s}{L}\cdot\left[\frac{L}{R} - \frac{L}{R}e^{-\frac{R}{L}t}\right]$$

$$= \frac{V_s}{R}\left[1 - e^{-\frac{R}{L}t}\right]$$

c) $i_c(t) = C\frac{d\,V_c(t)}{dt}$

$$I_c(s) = \mathcal{L}\left\{C\frac{d\,V_c(t)}{dt}\right\}$$

$$= CsV_c - C\,V_c(0)$$

d) $V_c = V_R = R\,i_c(t)$

$$V_c - R\,i_c(t) = 0$$

$$V_c - RC\frac{dV_c}{dt} = 0$$

$$\frac{V_c}{R} - C\frac{dV_c}{dt} = 0, \quad \frac{dV_c}{dt} = \frac{d}{dt}(R\cdot i_c(t))$$

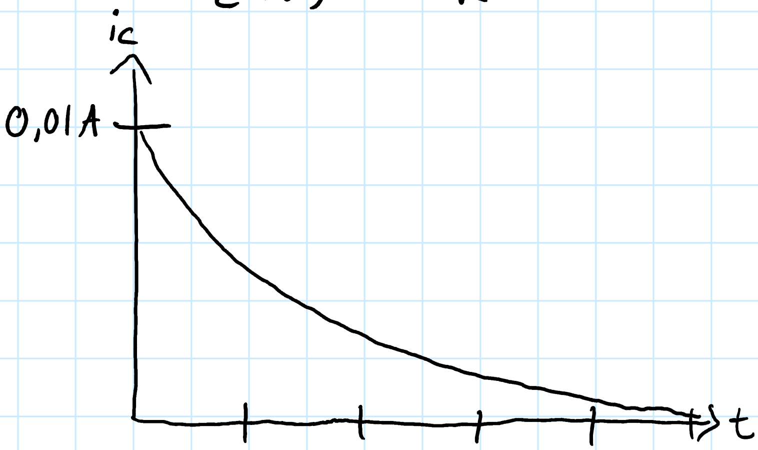
$$i(t) - RC\frac{d\,i_c(t)}{dt} = 0$$

e) $I(s) = RCs\,I(s) - RC\,i(0), \quad i(0) = \frac{V_0}{R}$

e) $I(s) = RCs\,I(s) - RC\,i(0)$, $\quad i(0) = \frac{V_0}{R}$

$$I(s) = \frac{-RC\frac{V_0}{R}}{1-RCs} = \frac{\frac{V_0}{R}}{s-\frac{1}{RC}}$$

$$i_c(t) = \frac{V_0}{R} e^{\frac{1}{RC}t}$$ Should be a minus



Problem 5

a) $$\frac{V_s}{R}\left[1 - e^{-\frac{R}{L}t}\right] = \frac{V_s}{R}\left(1 - \frac{1}{e}\right)$$

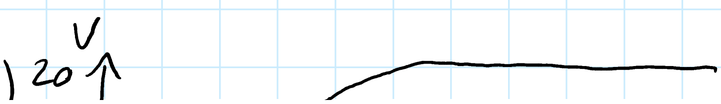
$$e^{-\frac{R}{L}t} = e^{-1} \Rightarrow t = \frac{L}{R} = \tau$$

b) $$i_c(t) = \frac{V_0}{R} e^{-\frac{t}{RC}}$$

$$i_c(\tau) = \frac{V_0}{R} e^{-\frac{\tau}{RC}} = \frac{V_0}{RC} \cdot 1 - \left(1 - \frac{1}{e}\right)$$

$$\Rightarrow e^{-\frac{\tau}{RC}} = e^{-1} \Rightarrow \tau = RC$$

c)



c)